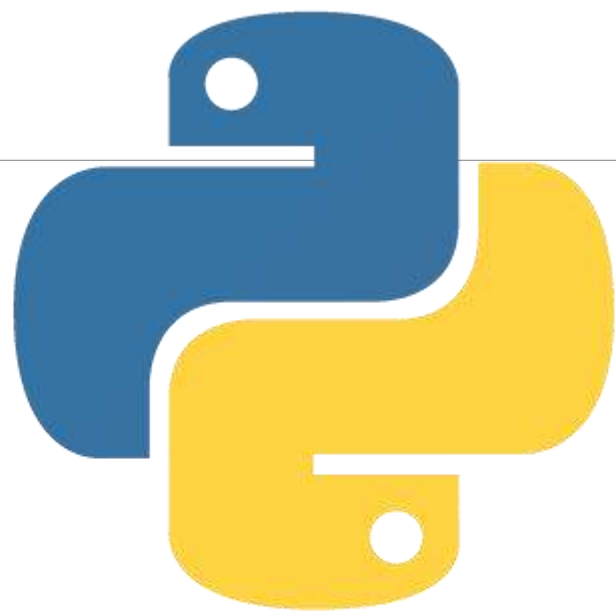


PYTHON

IT 5/L



python™

What is python programming language?

- Is a general purpose, high level, interpreted language with easy syntax and dynamic semantics.
- Python is a popular programming language.
- The world's fastest growing programming language.
- It was created by **Guido van Rossum**, and released in **1991**.
- The name Python is from Monty Python's Flying Circus.

It is used for:

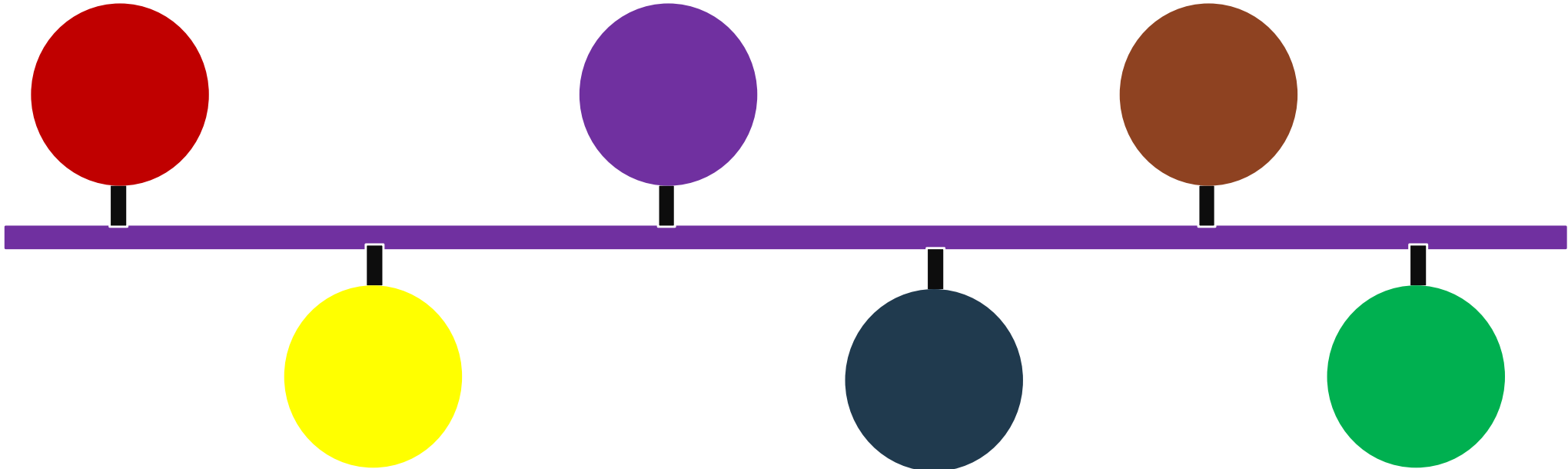
web development (server-side), software

development, mathematics, and system scripting.

Software Engineers

Data Analysts

Accountants



Mathematicians

Scientists

Network Engineers

What is python programming language?

- Can solve complex problems in less time with fewer lines of code.
- Multi-purpose language with a simple, and beginner-friendly syntax.

Example:

JAVA – `System.out.println(“Hello World”);`

PYTHON – `print(“Hello World”)`

What can you do with Python?

- ❖ Data Analysis
- ❖ Automation
- ❖ Artificial Intelligence
- ❖ Web, Mobile, Desktop Application
- ❖ Software testing
- ❖ Hacking



Instagram



Pinterest



Spotify



Dropbox



Uber

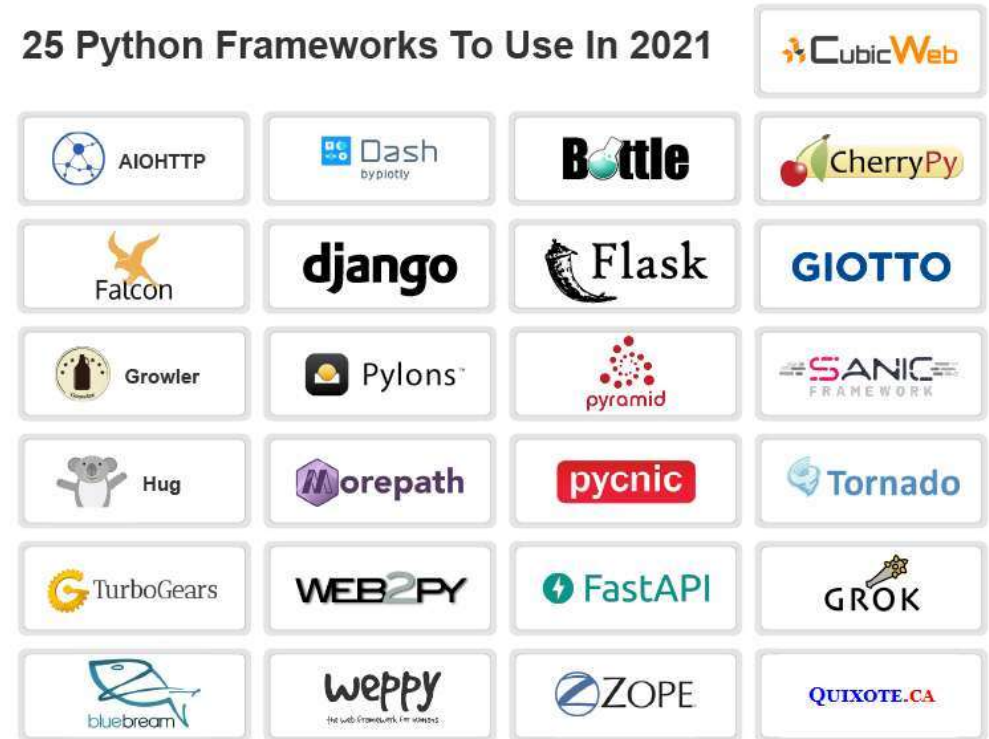


Reddit

Why python programming language?

- High-Level
- Cross-Platform
- Huge Community
- Large Ecosystem

25 Python Frameworks To Use In 2021



Why Python?

- Python works on different platforms (Windows, Mac, Linux, Raspberry Pi, etc).
- Python has a simple syntax similar to the English language. Python has syntax that allows developers to write programs with fewer lines than some other programming languages.
- Python runs on an interpreter system, meaning that code can be executed as soon as it is written. This means that prototyping can be very quick.
- Python can be treated in a procedural way, an object-oriented way or a functional way.

Advantage of using Python

- Presence of third-party modules
- Extensive support libraries(NumPy for numerical calculations, Pandas for data analytics, etc.)
- Open source and large active community base
- Versatile, Easy to read, learn and write
- User-friendly data structures
- High-level language

Advantage of using Python

- Dynamically typed language(No need to mention data type based on the value assigned, it takes data type)
- Object-Oriented and Procedural Programming language
- Portable and Interactive
- Ideal for prototypes – provide more functionality with less coding
- Highly Efficient(Python's clean object-oriented design provides enhanced process control, and the language is equipped with excellent text processing and integration capabilities, as well as its own unit testing framework, which makes it more efficient.)
- Internet of Things(IoT) Opportunities
- Interpreted Language
- Portable across Operating systems

Disadvantage of using Python

- **Performance:** Python is an interpreted language, which means that it can be slower than compiled languages like C or Java. This can be an issue for performance-intensive tasks.
- **Global Interpreter Lock:** The Global Interpreter Lock (GIL) is a mechanism in Python that prevents multiple threads from executing Python code at once. This can limit the parallelism and concurrency of some applications.
- **Memory consumption:** Python can consume a lot of memory, especially when working with large datasets or running complex algorithms.

Disadvantage of using Python

- Dynamically typed: Python is a dynamically typed language, which means that the types of variables can change at runtime. This can make it more difficult to catch errors and can lead to bugs.
- Packaging and versioning: Python has a large number of packages and libraries, which can sometimes lead to versioning issues and package conflicts.
- Lack of strictness: Python's flexibility can sometimes be a double-edged sword. While it can be great for rapid development and prototyping, it can also lead to code that is difficult to read and maintain.
- Steep learning curve: While Python is generally considered to be a relatively easy language to learn, it can still have a steep learning curve for beginners, especially if they have no prior experience with programming.

Python Syntax compared to other programming languages

Python was designed for readability, and has some similarities to the **English language** with influence from **mathematics**.

Python uses **new lines** to complete a command, as opposed to other programming languages which often use **semicolons** or **parentheses**.

Python relies on **indentation**, using **whitespace**, to **define scope**; such as the scope of **loops, functions and classes**. Other programming languages often use **curly-brackets** for this purpose.

Features of Python

- ❑ Simplicity
- ❑ Open Source
- ❑ Embeddable & Extensible
- ❑ Interpreted
- ❑ Huge Libraries
- ❑ Object Oriented

Industry that uses Python

- ❑ Google
- ❑ DropBox
- ❑ Netflix
- ❑ National Security
- ❑ BitTorrent
- ❑ NASA

Learning Path

- ☐ Variables, Data type, operators
- ☐ Arrays
- ☐ Flow Control
- ☐ Methods
- ☐ File handling
- ☐ OOP
- ☐ Practice

Software and Hardware Requirement

Python Install

- To install Python, go to <https://www.python.org/> and download python.
- Many PCs and Macs will have python already installed.
- To check if you have python installed on a Windows PC, search in the start bar for Python or run the following on the Command Line (cmd.exe): “*python –version*”

Python Install

- In this class, we are going to use VS code as our code editor.
- To install, go to <https://code.visualstudio.com/> and download the version for your operating system.